Exercise: Data Typing and Structuring

Explore data typing by cleaning and converting price fields from strings to floats, then learn data structuring by adding cumulative sum columns. This lesson guides you through preparing and enriching data for ETL transformations using pandas.

Data typing

Suppose we extracted a dataset that includes a field called `Price` that represents the price of a product in dollars. However, upon further inspection, we noticed that the column's data type is a string instead of a float. Furthermore, the prices are formatted with a dollar sign and comma, such as `'$1,000.00'`.

We know the data should eventually be loaded into a database table with a column data type of float. Our task is to remove the dollar sign and commas from the `Price` column and then convert the data type to `float`.

Let's look at the `Price` column using the `head()` method, which prints the first five values in the column. It also prints the column's data type, which is Object. The **Object** data type is used for strings in pandas.

main.py Python 3.10.4

```python
import pandas as pd

# Loading the Price column into a pandas DataFrame
df = pd.DataFrame({'Price': ['$800.30', '$6,692.37', '$1,523.00', '$1,22

# Print column
print(df.Price.head())
```

Ask

Run

Price column

Task 1: Remove the dollar signs and commas

Eliminate all the characters in the `Price` column and keep only the digits.

> **Note:** Fill in the missing code blocks to complete the tasks.

main.py Python 3.10.4

```
# Create a new list to store the cleaned prices
prices = []

# Iterate over the Price column in the DataFrame
for price in df.Price.values:

    # Remove the dollar sign and commas from each price
    <<InsertHere>>

    # Add the cleaned price to the new list
    <<InsertHere>>

# Replace the Price column with the cleaned prices
```

Run

Removing the dollar signs and commas

Ask

Task 2: Convert the `Price` column to float

After removing unnecessary characters, convert the `Price` column to float. Use the built-in pandas method `astype()` to accomplish that.

main.py Python 3.10.4

```
# Convert the Price column to a float
<<InsertHere>> = <<InsertHere>>.<<InsertHere>>("float")

# Print the Price column. It should be a float without the dollar sign an
print(df.Price.head())
```

Run

Converting the Price column to float

Solution code: Data typing

Here's the solution code for the above tasks:

main.py Python 3.10.4

```
import pandas as pd

# Loading the Price column into a pandas DataFrame
df = pd.DataFrame({'Price': ['$800.30', '$6,692.37', '$1,523.00', '$1,224

# Create a new list to store the cleaned prices
prices = []

# Iterate over the Price column in the DataFrame
for price in df.Price.values:

    # Remove the dollar sign and commas from each price
```

Ask

Run

Solution code: Data typing

- **Line 10:** We iterate over the `Price` column.
- **Line 13:** We replace the dollar sign and comma with an empty string for each price.
- **Line 16:** We append new and clean prices to a list called `prices`.
- **Line 19:** We replace the `Price` column with the list `prices`.
- **Line 22:** We convert the clean `price` column to float.

Data structuring

Next, modify the structure of the data by adding a new column and enriching the data. Add the column `Cumulative_Sum` to the DataFrame. This column will contain the cumulative sum of the `Price` column. This can be done this by using the `cumsum()` method on the `Price` column and assigning it to the new column.

Task 1: Creating the `Cumulative_Sum` column

main.py Python 3.10.4

```
# Add column "Cumulative_Sum"
<<InsetHere>>

# Print DataFrame
print(df)
```

Ask

Run

Creating the “Cumulative_Sum” column

Solution code

Here’s the solution code:

main.py Python 3.10.4

```
# Add column "Cumulative_Sum"
df["Cumulative_Sum"] = df.Price.cumsum()

# Print DataFrame
print(df)
```

Run

Solution code: Data structuring

- **Line 2:** We create a new column called `Cumulative_Sum` using the built-in pandas method `cumsum()`.

Ask